

VARUN CHANDRASHEKHAR GOHIL

Email: varuncg@mit.edu

Phone: (+1) 607-405-9483

Website: varungohil.github.io

EDUCATION

Massachusetts Institute of Technology	September 2022 - Present
PhD, Electrical Engineering and Computer Science	GPA: 5.0/5.0
Indian Institute of Technology Gandhinagar	July 2016 - July 2020
B.Tech, Computer Science and Engineering	CPI: 9.04/10

AWARDS AND HONORS

· Best Paper of Computer Architecture Letters	2024
· Best Presentation Award Finalist, HotStorage	2020
· Awarded the Jacobs Fellowship at Cornell University	2021
· Awarded cash prize of Rs.12,500 by IIT Gandhinagar for undergraduate research	2021
· Placed third in ArchReasoning Challenge held at ISCA	2025
· Received Special Mention in Undergraduate Research Conclave, IIT Gandhinagar	2019

PUBLICATIONS

When Machine Learning Isn't Sure: Building Resilient ML-Based Computer Systems by Embracing Uncertainty

[Varun Gohil](#), Nevena Stojkovic, Noman Bashir, Sundar Dev, Gaurang Upasani, David Lo, Parthasarathy Ranganathan, Christina Delimitrou

To appear in Annual Conference on Machine Learning and Systems, (**MLSys**), 2026

QuArch: A Benchmark for Evaluating LLM Reasoning in Computer Architecture

Shvetank Prakash, Andrew Cheng, Arya Tschand, Mark Mazumder, [Varun Gohil](#), Jeffrey Ma, Jason Yik, Zishen Wan, Jessica Quaye, Elisavet Lydia Alvanaki, Avinash Kumar, Chandrashis Mazumdar, Tuhin Khare, Alexander Ingare, Ikechukwu Uchendu, Radhika Ghosal, Abhishek Tyagi, Chenyu Wang, Andrea Mattia Garavagno, Sarah Gu, Alice Guo, Grace Hur, Luca Carloni, Tushar Krishna, Ankita Nayak, Amir Yazdanbakhsh, and Vijay Janapa Reddi

To appear in Forty-third International Conference on Machine Learning (**ICML**), 2026

The Case for Uncertainty-Governed Predictor Hierarchies in ML for Systems

Christos Zarkos, Nevena Stojkovic, [Varun Gohil](#), Christina Delimitrou

Workshop on AI for Computing Systems Design (**Architecture 2.0**) at ASPLOS 2026

The Importance of Generalizability in Machine Learning for Systems

[Varun Gohil](#), Sundar Dev, Gaurang Upasani, David Lo, Parthasarathy Ranganathan, Christina Delimitrou

IEEE Computer Architecture Letters, (**CAL**), 2024

Selected as Best of IEEE Computer Architecture Letters 2024

Sabre: Improving Memory Prefetching in Serverless MicroVMs with Near-Memory Hardware-Accelerated Compression

Nikita Lazarev, [Varun Gohil](#), James Tsai, Andy Anderson, Bhushan Chitlur, Zhiru Zhang, Christina Delimitrou

18th USENIX Symposium on Operating Systems Design and Implementation (**OSDI**), 2024

The Sunk Carbon Fallacy: Rethinking Carbon Footprint Metrics for Effective Carbon-Aware Scheduling

Noman Bashir, [Varun Gohil](#), Anagha Belavadi Subramanya, Mohammad Shahrad, David Irwin, Elsa Olivetti, and Christina Delimitrou

ACM Symposium on Cloud Computing, (**SoCC**), 2024

Performance optimization opportunities in the Android software stack

[Varun Gohil](#)*, Nisarg Ujjainkar*, Joycee Mekie, Manu Awasthi

BenchCouncil Transactions on Benchmarks, Standards and Evaluations (**TBench**), October 2021

Fixed-Posit: A Floating-Point Representation for Error-Resilient Applications

[Varun Gohil](#)*, Sumit Walia*, Joycee Mekie, Manu Awasthi

IEEE Transactions on Circuits and Systems II : Express Briefs (**TCAS-II**), April 2021

Prefetching in Hybrid Main Memory Systems

Subisha V, [Varun Gohil](#), Nisarg Ujjainkar, Manu Awasthi

12th USENIX Workshop on Hot Topics in Storage and File Systems (**HotStorage**) 2020

[Reproducibility Report] One ticket to win them all: generalizing lottery ticket initializations across datasets and optimizers

[Varun Gohil](#)*, S. Deepak Narayanan*, Atishay Jain*

NeurIPS Reproducibility Challenge, **ReScience C**, 2020.

Effect of Feature Hashing on Fair Classification

Ritik Dutta*, [Varun Gohil](#)*, Atishay Jain*

Young Researchers' Symposium, ACM India Joint International Conference on Data Science & Management of Data, (**CODS-COMADS**) 2020

FAB: Framework for Analyzing Benchmarks

[Varun Gohil](#)*, Shreyas Singh*, Manu Awasthi

Work in Progress Track, 10th International Conference on Performance Engineering (**ICPE**) 2019

META: Memory Exploration Tool for Android Devices

Nisarg Parikh, [Varun Gohil](#), Manu Awasthi

Poster Track, 24th International Conference on Mobile Computing and Networking (**MobiCom**) 2018

PRE-PRINTS

Benchmarking Compound AI Applications for Hardware-Software Co-Design

Paramuth Samuthrsindh, Angel Cervantes, [Varun Gohil](#), Gohar Irfan Chaudhry, Christina Delimitrou, Adam Belay

EXPERIENCE

Student Researcher, Google

June 2025 - August 2025

Hosts : Swinay Subramanian, Kavya Sreedhar, Amir Yazdanbakhsh

Part-time Student Researcher, Google

June 2024 - August 2024

Hosts : Sundar Dev, David Lo, Gaurang Upasani

Research Intern, Google

June 2023 - August 2023

Hosts : Sundar Dev, David Lo, Gaurang Upasani

Research Intern, Google

June 2022 - August 2022

Hosts : Sundar Dev, David Lo

Graduate Researcher, Cornell University

August 2021 - May 2022

Advisor : Prof. Christina Delimitrou

Research Fellow, Ashoka University

Advisor : Prof. Manu Awasthi

August 2020 - July 2021

Visiting Scholar, University of Utah

Advisor : Prof. Rajeev Balasubramanian

May 2019 - July 2019

Summer Research Intern, IIT Gandhinagar

Advisor : Prof. Manu Awasthi

May 2018 - July 2018

SERVICE

· **Program Committee, Reviewer:**

- ML for Systems workshop, NeurIPS 2025, 2024
- ML for Computer Architecture and Systems (MLArchSys) workshop, ISCA 2026

· **Mentoring:**

- Kartik Pingle MIT UROP 2026-Present
Project : Interpreting ML models to extract system insights
- Angel Cervantes MIT UROP 2025-Present
Project : Benchmarking Compound-AI applications
- Paramuth Samuthrsindh MIT UROP 2025-Present
Project : Benchmarking Compound-AI applications
- Nevena Stojkovic MIT UROP 2025-26
Project : Uncertainty-aware resource management for microservices
Next: PhD student at Stanford
- Alan Song MIT PRIMES 2023
Project : Reinforcement learning approaches for serverless autoscaling
Selected as Regeneron Science Talent Search (STS) scholar
Next: Undergraduate student at UCLA
- Evan Ning MIT PRIMES 2023
Project : Reinforcement learning approaches for serverless autoscaling
Next: Undergraduate student at CMU

· **Teaching Assistant:**

- 6.S984 Datacenter Computing, MIT Spring 2026
- 6.1910 Computation Structures, MIT Fall 2025, 2024
- Machine Learning, IIT Gandhinagar Spring 2020
- Operating Systems, IIT Gandhinagar Fall 2019

· **Organization:**

- Grad School Application Tweet Storm with Students@Systems 2022
- Pre-conference volunteer, ICLR 2020
- Tutorial on DRAM timing parameters, Computer Architecture Winter School (CAWS) 2020